

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	AIML	Date:	

Code of Conduct

I N.Adi Narayana(192525346) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Adi

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science drives}) = 1$
- $C(\text{science drives innovation}) = 1$

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Questions

1. **Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} | \text{data})$. Interpret how this probability influences next-word prediction in a mobile keyboard application.**

For a Bigram Language Model, the Maximum Likelihood Estimate is:

$$P(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$$

Therefore,

$$P(\text{science} | \text{data}) = \frac{C(\text{data science})}{C(\text{data})}$$

Substituting the given values:

$$P(\text{science} | \text{data}) = \frac{3}{3}$$

$$P(\text{science} | \text{data}) = 1.0$$

Interpretation

The probability is 1.0 or 100% because every occurrence of data in the training corpus is followed by science.:

Data

the keyboard has very high confidence that the next word is:

science

In a mobile keyboard

This helps the predictive keyboard rank **science** as the most likely next word. A high conditional probability means the language model has strong evidence from the training corpus.

Previous word	Next word	Probability
data	science	1.00

2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.

The user enters:

Data science improves

The problem is that "**science improves**" and "**improves**" are unseen in the training corpus.

A Backoff Language Model uses a higher-order model when possible and falls back to lower-order models when the required n-gram is unavailable.

Step 1: Check Trigram

We would need:

$P(\text{improves}|\text{data},\text{science})$

The trigram:

data science improves

does not occur in the corpus.

Therefore:

$P(\text{improves}|\text{data},\text{science})=0$

under the unsmoothed MLE model.

Step 2: Back off to Bigram

We then check:

$P(\text{improves}|\text{science})$

The Bigram:

science improves

also does not occur.

Therefore:

$P(\text{improves}|\text{science})=0$

Step 3: Back off to Unigram

Finally, we check:

$P(\text{improves})$

The word **improves** does not occur anywhere in the corpus.

Therefore:

$P(\text{improves})=0$

Result

Thus, with the **given unsmoothed corpus**:

$P(\text{data science improves})=0$

because the final word is completely unseen.

Why is Backoff important?

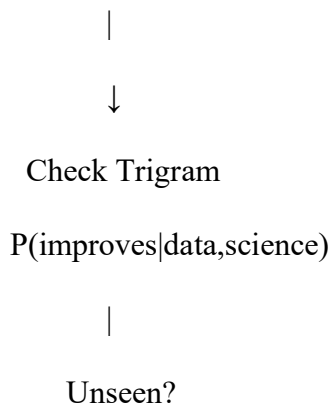
In a real mobile keyboard, users constantly type:

- new words
- names
- abbreviations
- slang
- technical terms
- previously unseen word combinations

If the system relied only on Trigrams, an unseen sequence would receive probability zero.

The Backoff strategy is:

Input: data science improves



In practical systems, **smoothing or an unknown-word probability** would normally be added so that an unseen word does not necessarily receive exactly zero probability.

Importance for real-time prediction

Backoff provides **robustness and coverage**. It allows a keyboard to continue making predictions even when a complete Trigram is unavailable.

3. **Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.**

Deleted Interpolation combines:

1. Trigram probability
2. Bigram probability
3. Unigram probability

The general formula is:

$$P(w_i | w_{i-2}, w_{i-1}) = \lambda_1 P_{tri} + \lambda_2 P_{bi} + \lambda_3 P_{uni}$$

Given:

$$\lambda_1 = 0.5, \quad \lambda_2 = 0.3, \quad \lambda_3 = 0.2$$

We want:

$$P(is | data, science)$$

Step 1: Trigram probability

The trigram is:

data science is

It occurs once.

$$C(data, science, is) = 1$$

and:

$$C(data, science) = 3$$

Therefore:

$$P_{tri}(is|data, science) = \frac{1}{3} = 0.3333$$

Step 2: Bigram probability

We need:

$$P(is|science)$$

Given:

$$C(science, is) = 2$$

The word **science** occurs 3 times:

$$C(science) = 3$$

Therefore:

$$P_{bi}(is|science) = \frac{2}{3} = 0.6667$$

Step 3: Unigram probability

The corpus contains 12 words:

Word	Count
data	3
science	3
is	2
powerful	1

drives	1
innovation	1
evolving	1
Total	12

Therefore:

$$P_{uni}(is) = \frac{2}{12} = \frac{1}{6} = 0.1667$$

Step 4: Apply interpolation

$$\begin{aligned} P(is|data, science) &= (0.5)(0.3333) + (0.3)(0.6667) + (0.2)(0.1667) \\ &= 0.1667 + 0.2000 + 0.0333 \end{aligned}$$

$$P(is|data, science) \approx 0.40$$

So the interpolated probability is approximately:

40%

Probability of the complete sequence

If the question interprets "probability of the sequence" as the conditional prediction of **is after data science**, the answer is:

$$P(is|data, science) = 0.40$$

If the complete sentence probability is required:

$$P(data, science, is) = P(data) \times P(science|data) \times P(is|data, science)$$

Since:

$$P(data) = \frac{3}{12} = 0.25$$

and:

$$P(science|data) = 1$$

we get:

$$P(data, science, is) = 0.25 \times 1 \times 0.40$$

$$P(data, science, is) = 0.10$$

Why interpolation is useful

Interpolation combines information from different levels:

Model	Probability	Weight
Trigram	0.3333	0.5
Bigram	0.6667	0.3
Unigram	0.1667	0.2
Interpolated	0.4000	

The **Trigram** provides context-specific information, the **Bigram** provides broader contextual information, and the **Unigram** provides general word frequency.

Thus, interpolation balances:

- **Reliability** → higher-order n-grams capture context.
- **Coverage** → lower-order n-grams help when detailed context is sparse.
- **Robustness** → predictions do not depend entirely on one model.

Note: Strict deleted interpolation normally estimates the interpolation weights using held-out/deleted counts. Since the question explicitly supplies $\lambda_1, \lambda_2, \lambda_3$, we use those given weights directly.

4. Calculate the Entropy of the distribution $P(is)=0.66$ and $P(drives)=0.33$. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

Entropy measures the **uncertainty** in a probability distribution.

The formula is:

$$H(X) = - \sum_i P(x_i) \log_2 P(x_i)$$

Therefore:

$$H(X) = -[P(is) \log_2 P(is) + P(drives) \log_2 P(drives)]$$

Substitute:

$$H(X) = -[0.66 \log_2(0.66) + 0.33 \log_2(0.33)]$$

Approximately:

$$H(X) = -[0.66(-0.599) + 0.33(-1.599)]$$

$$H(X) \approx 0.923$$

Therefore:

$H(X) \approx 0.92 \text{ bits}$

Important note

The supplied probabilities are:

$$0.66 + 0.33 = 0.99$$

rather than exactly 1.00 because of rounding. If normalized first:

$$P(is) = \frac{0.66}{0.99} = 0.6667$$

$$P(drives) = \frac{0.33}{0.99} = 0.3333$$

then:

$$H(X) \approx 0.918 \text{ bits}$$

So **about 0.92 bits** is the appropriate result.

Interpretation of Entropy

There are two possible next words:

- **is** → 66%
- **drives** → 33%

The keyboard has a preference for **is**, but it is not completely certain.

The entropy of approximately **0.92 bits** indicates a **moderate level of uncertainty**.

In an intelligent keyboard

Entropy	Interpretation
Low entropy	One prediction dominates → high confidence
Medium entropy	Several predictions are plausible
High entropy	Many predictions have similar probabilities → high uncertainty

For example:

$$P(is) = 0.99, \quad P(drives) = 0.01$$

would have very low entropy, meaning the keyboard is highly confident.

Whereas:

$$P(is) = 0.50, \quad P(drives) = 0.50$$

would have maximum entropy for two possible outcomes:

$$H = 1 \text{ bit}$$

Thus, entropy helps the keyboard determine **how confident or uncertain its next-word prediction is**.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- $P(\text{book} \mid \text{VB}) = 0.6$
- $P(\text{book} \mid \text{NN}) = 0.4$
- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{Start} \rightarrow \text{NN}) = 0.5$

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.

Sentence 1: "Book a flight ticket now."

Word	POS Tag	Meaning
Book	VB	Verb
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

Sentence 2: "This book is interesting."

Word	POS Tag	Meaning
This	DT	Determiner
book	NN	Noun
is	VBZ	Verb
interesting	JJ	Adjective

Justification: In "Book a flight," book is an action, so it is VB. In "This book," book refers to an object, so it is NN.

2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.

Given:

$$P(\text{book}|\text{VB}) = 0.6$$

$$P(\text{Start} \rightarrow \text{VB}) = 0.5$$

Therefore:

$$P(\text{VB}, \text{book}) = 0.5 \times 0.6$$

$$P(\text{VB}, \text{book}) = 0.30$$

For NN:

$$P(\text{NN}, \text{book}) = 0.5 \times 0.4 = 0.20$$

Since:

$$0.30 > 0.20$$

the HMM favors VB.

Reason: "Book a flight ticket now" describes an action, so **book** is naturally interpreted as a verb.

3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.

Rule-Based	HMM/Stochastic
Uses manually written rules	Uses probabilities from training data
Simple and predictable	Handles ambiguity better
Difficult to maintain for large systems	Adapts better to language patterns
Limited context handling	Uses contextual probabilities

Recommendation: HMM tagging is more suitable for a large-scale chatbot because it can handle ambiguous words such as "book" based on context and learned probabilities.

4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

Standardized POS tagsets such as the Penn Treebank Tagset help the chatbot understand grammatical structure.

They improve:

- Intent Detection: Identify actions such as *book*, *cancel*, and *change*.
- Response Generation: Understand whether a word represents an action, object, or description.
- Context Understanding: Resolve ambiguous words like *book*.
- Overall Performance: Provide consistent linguistic information for downstream NLP tasks.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. **Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.**

Initial tags:

economic/JJ growth/NN increases/NNS employment/NN

Rule:

Change NNS → VBZ if the preceding word is tagged NN.

Since **growth** = NN and **increases** = NNS, apply the rule:

economic/JJ growth/NN increases/VBZ employment/NN

Why? "Increases" is the main verb of the sentence, and VBZ is the correct tag for a third-person singular present-tense verb.

2. **Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.**

The main error is:

- increases/NNS ❌
- increases/VBZ ✅

The sentence structure is:

Subject + Verb + Object

- economic/JJ → describes growth.
- growth/NN → noun and subject.
- increases/VBZ → verb showing the action.
- employment/NN → noun/object.

Therefore, the corrected sequence is:

JJ NN VBZ NN

3. Assuming the corpus contains the words:

- **economic (120 occurrences)**
- **growth (450 occurrences)**
- **increases (210 occurrences)**
- **employment (380 occurrences)**

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

Total occurrences:

$$120+450+210+380=1160$$

Word	Frequency	Probability
economic	120	$120/1160=0.1034$
growth	450	$450/1160=0.3879$
increases	210	$210/1160=0.1810$
employment	380	$380/1160=0.3276$
Total	1160	1.0000

So the frequency distribution is approximately:

- **economic:** 10.34%
- **growth:** 38.79%
- **increases:** 18.10%
- **employment:** 32.76%

Importance: Frequent words provide stronger statistical evidence for probabilistic tagging and help the system learn likely word/tag patterns.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

Transformation-Based Tagging : starts with an initial POS assignment and applies learned rules to correct common errors.

Here:

NNS→VBZ

corrects **increases**.

Before correction

The tagger is uncertain because **increases** could potentially be interpreted as a noun or verb.

After correction

The grammatical context:

growth increases employment

strongly indicates that increases is a verb.

Therefore, uncertainty decreases.

Entropy is calculated as:

$$H = - \sum P(tag) \log_2 P(tag)$$

- **High entropy** → high uncertainty about the POS tag.
- **Low entropy** → high confidence in the POS tag.

Thus, the Brill Tagger improves accuracy by using **context-based transformation rules**, while entropy can measure whether confidence in the POS assignment improves after applying the rule.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____